

WATER POLLUTION AND RIVER CLEANING MISSIONS

SOURCE-PATHWAY MAP → STANDARD FIREWALL → ORGANIC-POLLUTION DIAGNOSTIC → EUTROPHICATION CHAIN → INSTITUTION AND APPROVAL LADDER → SEWAGE QUANTITY LEDGER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

SOURCE-PATHWAY MAP

SOURCE-PATHWAY MAP

POINT SOURCE

IDENTIFIABLE PIPE, DRAIN OR OUTFALL

NON-POINT SOURCE

DIFFUSE CATCHMENT RUNOFF

SOURCE TO RECEIVING WATER

SOURCE SAMPLE VERSUS CATCHMENT EVIDENCE

TOOL MUST MATCH THE PATHWAY

CORE CLAIM

POINT SOURCE → identifiable pipe, drain or outfall

NON-POINT SOURCE → diffuse catchment runoff

MECHANISM

PATHWAY → source to receiving water

MONITORING → source sample versus catchment evidence

CONSEQUENCE / LIMIT

CONTROL → tool must match the pathway

MUST REMEMBER: Water pollution links pollutant load, concentration, dissolved oxygen,...

CORE CLAIM

• POINT SOURCE → identifiable pipe, drain or outfall

• NON-POINT SOURCE → diffuse catchment runoff

MECHANISM

• PATHWAY → source to receiving water

• MONITORING → source sample versus catchment evidence

CONSEQUENCE / LIMIT

• CONTROL → tool must match the pathway

• MUST REMEMBER: Water pollution links pollutant load, concentration, dissolved oxygen,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Water pollution links pollutant load, concentration, dissolved oxygen,.

01

STAGE 01

STANDARD FIREWALL

STANDARD FIREWALL

EFFLUENT STANDARD

CONTROLS DISCHARGE FROM A SOURCE

WATER-QUALITY CRITERION

DESCRIBES RECEIVING WATER

USE CLASS

INTENDED-WATER-USE FRAMEWORK

UNIT AND SAMPLING

CORE CLAIM

EFFLUENT STANDARD → controls discharge from a source

WATER-QUALITY CRITERION → describes receiving water

MECHANISM

USE CLASS → intended-water-use framework

UNIT AND SAMPLING → belong to the exact indicator

CONSEQUENCE / LIMIT

RULE → never exchange these values

EFFLUENT STANDARD → controls discharge from a source

CORE CLAIM

• EFFLUENT STANDARD → controls discharge from a source

• WATER-QUALITY CRITERION → describes receiving water

MECHANISM

• USE CLASS → intended-water-use framework

• UNIT AND SAMPLING → belong to the exact indicator

CONSEQUENCE / LIMIT

• RULE → never exchange these values

• EFFLUENT STANDARD → controls discharge from a source

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UNIT AND SAMPLING → belong to the exact indicator.

02

STAGE 02

ORGANIC-POLLUTION DIAGNOSTIC

ORGANIC-POLLUTION DIAGNOSTIC

ORGANIC LOAD

MICROBIAL DECOMPOSITION

OXYGEN DEMAND DURING DECOMPOSITION

OXYGEN REMAINING FOR AQUATIC LIFE

FAECAL INDICATOR

CONTAMINATION WARNING

USE INDICATORS TOGETHER, NOT INTERCHANGEABLY

CORE CLAIM

ORGANIC LOAD → microbial decomposition

BOD → oxygen demand during decomposition

MECHANISM

DO → oxygen remaining for aquatic life

FAECAL INDICATOR → contamination warning

CONSEQUENCE / LIMIT

DIAGNOSIS → use indicators together, not interchangeably

ORGANIC LOAD → microbial decomposition

CORE CLAIM

• ORGANIC LOAD → microbial decomposition

• BOD → oxygen demand during decomposition

MECHANISM

• DO → oxygen remaining for aquatic life

• FAECAL INDICATOR → contamination warning

CONSEQUENCE / LIMIT

• DIAGNOSIS → use indicators together, not interchangeably

• ORGANIC LOAD → microbial decomposition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAECAL INDICATOR → contamination warning; DIAGNOSIS → use indicators together, not interchangeably.

03

STAGE 03

EUTROPHICATION CHAIN

EUTROPHICATION CHAIN

NUTRIENT INPUT

NITROGEN OR PHOSPHORUS ENRICHMENT

ALGAL GROWTH

BIOMASS INCREASE

OXYGEN DEMAND

LOW OXYGEN

ECOLOGICAL STRESS

NUTRIENT INPUT → nitrogen or phosphorus enrichment

ALGAL GROWTH → biomass increase

DECOMPOSITION → oxygen demand

LOW OXYGEN → ecological stress

BOUNDARY → not identical to toxic industrial discharge

CORE CLAIM

MECHANISM

CONSEQUENCE / LIMIT

ORGANIC LOAD → microbial decomposition

BOD → oxygen demand during decomposition

DO → oxygen remaining for aquatic life

FAECAL INDICATOR → contamination warning

DIAGNOSIS → use indicators together, not interchangeably

ORGANIC LOAD → microbial decomposition

CORE CLAIM

- ORGANIC LOAD → microbial decomposition
- BOD → oxygen demand during decomposition

MECHANISM

- DO → oxygen remaining for aquatic life
- FAECAL INDICATOR → contamination warning

CONSEQUENCE / LIMIT

- DIAGNOSIS → use indicators together, not interchangeably
- ORGANIC LOAD → microbial decomposition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAECAL INDICATOR → contamination warning; DIAGNOSIS → use indicators together, not interchangeably.

03

STAGE 03

EUTROPHICATION CHAIN

EUTROPHICATION CHAIN

NUTRIENT INPUT

NITROGEN OR PHOSPHORUS ENRICHMENT

ALGAL GROWTH

BIOMASS INCREASE

OXYGEN DEMAND

LOW OXYGEN

ECOLOGICAL STRESS

NUTRIENT INPUT → nitrogen or phosphorus enrichment

→

ALGAL GROWTH → biomass increase

→

DECOMPOSITION → oxygen demand

→

LOW OXYGEN → ecological stress

→

BOUNDARY → not identical to toxic industrial discharge

CORE CLAIM

- NUTRIENT INPUT → nitrogen or phosphorus enrichment
- ALGAL GROWTH → biomass increase

MECHANISM

- DECOMPOSITION → oxygen demand
- LOW OXYGEN → ecological stress

CONSEQUENCE / LIMIT

- BOUNDARY → not identical to toxic industrial discharge
- NUTRIENT INPUT → nitrogen or phosphorus enrichment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOUNDARY → not identical to toxic industrial discharge.

04

STAGE 04

INSTITUTION AND APPROVAL LADDER

INSTITUTION AND APPROVAL LADDER

WATER ACT

STATUTORY POLLUTION-CONTROL FRAMEWORK

NATIONAL COORDINATION AND FRAMEWORK

SPCB OR PCC

CONSENT, MONITORING AND ENFORCEMENT

CTE OR CTO

SOURCE PERMISSION LAYER

WATER ACT → statutory pollution-control framework

→

CPCB → national coordination and framework

→

SPCB OR PCC → consent, monitoring and enforcement

→

CTE OR CTO → source permission layer

CORE CLAIM

- WATER ACT → statutory pollution-control framework
- CPCB → national coordination and framework

MECHANISM

- SPCB OR PCC → consent, monitoring and enforcement
- CTE OR CTO → source permission layer

CONSEQUENCE / LIMIT

- ENVIRONMENTAL CLEARANCE → separate prior-appraisal layer
- WATER ACT → statutory pollution-control framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPCB OR PCC → consent, monitoring and enforcement.

05

STAGE 05

SEWAGE QUANTITY LEDGER

SEWAGE QUANTITY LEDGER

TOTAL WASTEWATER PRODUCED

FLOW REACHING A NETWORK

NAMEPLATE TREATMENT CAPACITY

ACTUALLY TREATED

OPERATING INFLOW AND PERFORMANCE

REUSED OR DISCHARGED

FINAL PATHWAY AND QUALITY

GENERATED → total wastewater produced

→

SEWERED → flow reaching a network

→

INSTALLED → nameplate treatment capacity

→

ACTUALLY TREATED → operating inflow and performance

→

REUSED OR DISCHARGED → final pathway and quality

CORE CLAIM

- GENERATED → total wastewater produced
- SEWERED → flow reaching a network

MECHANISM

- INSTALLED → nameplate treatment capacity
- ACTUALLY TREATED → operating inflow and performance

CONSEQUENCE / LIMIT

- REUSED OR DISCHARGED → final pathway and quality
- CLOSE DISTINCTION: BOD is not COD, sewage generation is not treatment capacity or actual...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: BOD is not COD, sewage generation is not treatment capacity or actual.

06

STAGE 06

TREATMENT-PLANT MATRIX

TREATMENT-PLANT MATRIX

MUNICIPAL OR DOMESTIC SEWAGE

ONE ESTABLISHMENT'S INDUSTRIAL EFFLUENT

COMMON TREATMENT FOR A GROUP OF UNITS

INFRASTRUCTURE OUTPUT

COMPLIANT OPERATION

SEPARATELY VERIFIED PERFORMANCE

CORE CLAIM

STP → municipal or domestic sewage

ETP → one establishment's industrial effluent

MECHANISM

CETP → common treatment for a group of units

INSTALLATION → infrastructure output

CONSEQUENCE / LIMIT

COMPLIANT OPERATION → separately verified performance

STP → municipal or domestic sewage

CORE CLAIM

- STP → municipal or domestic sewage

MECHANISM

- CETP → common treatment for a group of units

CONSEQUENCE / LIMIT

- COMPLIANT OPERATION → separately verified performance

06

CORE CLAIM

- GENERATED → total wastewater produced
- SEWERED → flow reaching a network

MECHANISM

- INSTALLED → nameplate treatment capacity
- ACTUALLY TREATED → operating inflow and performance

CONSEQUENCE / LIMIT

- REUSED OR DISCHARGED → final pathway and quality
- CLOSE DISTINCTION: BOD is not COD, sewage generation is not treatment capacity or actual...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: BOD is not COD, sewage generation is not treatment capacity or actual.

06

STAGE 06

TREATMENT-PLANT MATRIX

TREATMENT-PLANT MATRIX

MUNICIPAL OR DOMESTIC SEWAGE

ONE ESTABLISHMENT'S INDUSTRIAL EFFLUENT

COMMON TREATMENT FOR A GROUP OF UNITS

INFRASTRUCTURE OUTPUT

COMPLIANT OPERATION

SEPARATELY VERIFIED PERFORMANCE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
STP → municipal or domestic sewage	CETP → common treatment for a group of units	COMPLIANT OPERATION → separately verified performance
ETP → one establishment's industrial effluent	INSTALLATION → infrastructure output	STP → municipal or domestic sewage

CORE CLAIM

- STP → municipal or domestic sewage
- ETP → one establishment's industrial effluent

MECHANISM

- CETP → common treatment for a group of units
- INSTALLATION → infrastructure output

CONSEQUENCE / LIMIT

- COMPLIANT OPERATION → separately verified performance
- STP → municipal or domestic sewage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CETP → common treatment for a group of units.

07

STAGE 07

REGULATOR-MISSION FIREWALL

REGULATOR-MISSION FIREWALL

CPCB OR SPCB

STATUTORY POLLUTION-CONTROL ROLE

RIVER MISSION

PROJECT AND BASIN COORDINATION

GANGA MISSION IMPLEMENTATION ARCHITECTURE

SEWERAGE SERVICE AND LOCAL OPERATION

ONE ACTOR NEVER SUBSTITUTES FOR ALL OTHERS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CPCB OR SPCB → statutory pollution-control role	NMCG → Ganga mission implementation architecture	RULE → one actor never substitutes for all others
RIVER MISSION → project and basin coordination	ULB → sewerage service and local operation	CPCB OR SPCB → statutory pollution-control role

CORE CLAIM

- CPCB OR SPCB → statutory pollution-control role
- RIVER MISSION → project and basin coordination

MECHANISM

- NMCG → Ganga mission implementation architecture
- ULB → sewerage service and local operation

CONSEQUENCE / LIMIT

- RULE → one actor never substitutes for all others
- CPCB OR SPCB → statutory pollution-control role

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → one actor never substitutes for all others.

08

STAGE 08

RIVER-MISSION COMPONENT WHEEL

RIVER-MISSION COMPONENT WHEEL

SEWAGE INFRASTRUCTURE

INTERCEPT AND TREAT FLOW

INDUSTRIAL MONITORING

SOURCE COMPLIANCE

BIODIVERSITY AND RIVERBANK MEASURES

PUBLIC PARTICIPATION

BEHAVIOUR AND ACCOUNTABILITY

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SEWAGE INFRASTRUCTURE → intercept and treat flow	ECOLOGY → biodiversity and riverbank measures	VISIBLE WORK → not itself a water-quality outcome
INDUSTRIAL MONITORING → source compliance	PUBLIC PARTICIPATION → behaviour and accountability	SEWAGE INFRASTRUCTURE → intercept and treat flow

CORE CLAIM

- SEWAGE INFRASTRUCTURE → intercept and treat flow
- INDUSTRIAL MONITORING → source compliance

MECHANISM

- ECOLOGY → biodiversity and riverbank measures
- PUBLIC PARTICIPATION → behaviour and accountability

CONSEQUENCE / LIMIT

- VISIBLE WORK → not itself a water-quality outcome
- SEWAGE INFRASTRUCTURE → intercept and treat flow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VISIBLE WORK → not itself a water-quality outcome.

09

STAGE 09

PROGRAMME-SCOPE GATE

PROGRAMME-SCOPE GATE

GANGA QUESTION

USE NOTIFIED GANGA ARCHITECTURE

OTHER RIVER

IDENTIFY APPLICABLE PROGRAMME AND STATE ACTORS

WIDER RIVER-CONSERVATION PROGRAMME CONTEXT

RETAIN RIVER, STRETCH AND DATE

NO TRANSFER

GANGA QUESTION → use notified Ganga architecture

→

OTHER RIVER → identify applicable programme and state actors

→

NRCP → wider river-conservation programme context

→

RESULT → retain river, stretch and date

→

NO TRANSFER → never move one mission result to another river

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
GANGA QUESTION → use notified Ganga architecture	NRCP → wider river-conservation programme context	NO TRANSFER → never move one mission result to another river
OTHER RIVER → identify applicable programme and state actors	RESULT → retain river, stretch and date	GANGA QUESTION → use notified Ganga architecture

CORE CLAIM

- GANGA QUESTION → use notified Ganga architecture
- OTHER RIVER → identify applicable programme and state actors

MECHANISM

- NRCP → wider river-conservation programme context
- RESULT → retain river, stretch and date

CONSEQUENCE / LIMIT

- NO TRANSFER → never move one mission result to another river
- GANGA QUESTION → use notified Ganga architecture

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO TRANSFER → never move one mission result to another river.

WATER POLLUTION AND RIVER CLEANING MISSIONS • TILE 3/4 • same master rows 4693-7927 of 10274 • continues below

